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Problem Set 1

Probability Models, Distributions, Joint Distributions, Transformations, and Order Statistics

Problem 1. Constructing a Distribution from Constraints [12 points]

A continuous random variable X has probability density function

$$f_X(x) = \begin{cases} a + bx, & 0 < x < 2, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\mathbb{E}(X) = \frac{7}{6}.$$

(a) Using only the defining properties of a probability density function and the given expectation, derive a system of equations for a and b.

Answer:

For the first equation, By the definition of a valid probability density function (PDF),

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

Since $f_X(x) = a + bx$ for $0 < x < 2$ and 0 otherwise, then

$$\begin{aligned} \int_0^2 (a + bx)\,dx = 1 &\implies \left[ax + \frac{b}{2}x^2\right]_{x=0}^{x=2} = 1 \\ &\implies a(2) + \frac{b}{2}(2)^2 - a(0) - \frac{b}{2}(0)^2 = 1 \\ &\implies 2a + 2b = 1. \end{aligned}$$

For the second equation,

By the definition of expectation,

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx = \frac{7}{6}$$

Then,

$$\begin{aligned}\int_0^2 x(a+bx)\,dx = \frac{7}{6} &\implies \left[\frac{ax^2}{2} + \frac{bx^3}{3}\right]_{x=0}^{x=2} = \frac{7}{6} \\ &\implies \frac{a(2)^2}{2} + \frac{b(2)^3}{3} - \frac{a(0)^2}{2} - \frac{b(0)^3}{3} = \frac{7}{6} \\ &\implies \frac{4a}{2} + \frac{8b}{3} = \frac{7}{6} \\ &\implies 2a + \frac{8b}{3} = \frac{7}{6}.\end{aligned}$$

(b) Solve the system and determine the values of a and b.

(c) Prove that the resulting function is a valid probability density function on its stated support.

Answer:
With $a = \frac{1}{4}$ and $b = \frac{1}{4}$, the resulting probability density function is

$$f_X(x) = \begin{cases} \frac{1}{4} + \frac{1}{4}x = \frac{1}{4}(1+x), & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

To prove that $f_X(x)$ is a valid probability density function, we verify the two required axioms:

1. **Non-negativity:** $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
2. **Normalization:** $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$.

For the Non-negativity condition ($f_X(x) \geq 0$): For $x \in (0,2)$: Since $x > 0$, we have $1 + x > 1 > 0$. Thus,

$$f_X(x) = \frac{1}{4}(1+x) > 0.$$

For $x \notin (0,2)$:

$$f_X(x) = 0 \geq 0.$$

Thus, $f_X(x) \geq 0$ holds for all $x \in \mathbb{R}$.

For the Normalization condition ($\int_{-\infty}^{\infty} f_X(x)\,dx = 1$):

Evaluating the total area under $f_X(x)$,

$$\begin{aligned}\int_{-\infty}^{\infty} f_X(x)\,dx &= \int_0^2 \left(\frac{1}{4}+\frac{1}{4}x\right) dx \\ &= \left[\frac{1}{4}x+\frac{1}{8}x^2\right]_0^2 \\ &= \left(\frac{1}{4}(2)+\frac{1}{8}(2)^2\right) - \left(\frac{1}{4}(0)+\frac{1}{8}(0)^2\right) \\ &= \left(\frac{1}{2}+\frac{4}{8}\right) - 0 \\ &= \frac{1}{2}+\frac{1}{2} \\ &= 1.\end{aligned}$$

Since both conditions—non-negativity ($f_X(x) \geq 0$ for all $x \in \mathbb{R}$) and normalization ($\int_{-\infty}^{\infty} f_X(x)\,dx = 1$)—are satisfied, $f_X(x)$ is formally proven to be a valid probability density function on its stated support.

(d) Derive the complete piecewise cumulative distribution function $F_X(x)$.

(e) Derive $\mathbb{E}(X^2)$ and $\mathrm{Var}(X)$.

(f) Let m denote the median of X. Determine m exactly and decide whether m is less than, equal to, or greater than $\mathbb{E}(X)$. Justify your conclusion analytically.

Problem 2. Distribution Identification Through Moment-Generating Functions [12 points]

Let $X_1, X_2, \ldots, X_n$ be independent Bernoulli random variables satisfying

$$P(X_i = 1) = p_i, \quad P(X_i = 0) = 1 - p_i, \quad i = 1, \ldots, n,$$

where the probabilities $p_1, \ldots, p_n$ are not assumed to be equal. Define

$$S_n = X_1 + X_2 + \cdots + X_n.$$

(a) Derive $M_{X_i}(t)$ directly from the definition of the moment-generating function.

Answer:
Since X_i is a Bernoulli random variable, it only takes values $x \in \{0, 1\}$

$$P(X_i = 1) = p_i \quad \text{and} \quad P(X_i = 0) = 1 - p_i.$$

By definition, the moment-generating function for a random variable is

$$M_{X_i}(t) = \mathbb{E}\left[e^{tX_i}\right].$$

Then, by the definition of expectation,

$$\begin{aligned} M_{X_i}(t) &= \sum_{x\in\{0,1\}} e^{tx} P(X_i = x) \\ &= e^{t(0)} P(X_i = 0) + e^{t(1)} P(X_i = 1) \\ &= 1 \cdot (1 - p_i) + e^t p_i \\ M_{X_i}(t) &= (1 - p_i) + p_i e^t. \end{aligned}$$

(b) Using independence, derive $M_{S_n}(t)$.

Answer:
By the definition of moment-generating function, the moment-generating function of the sum S_n is

$$M_{S_n}(t) = \mathbb{E}\left[e^{tS_n}\right].$$

Since $S_n = X_1 + X_2 + \cdots + X_n$, then

$$M_{S_n}(t) = \mathbb{E}\left[e^{t(X_1+X_2+\cdots+X_n)}\right] = \mathbb{E}\left[e^{tX_1} e^{tX_2} \cdots e^{tX_n}\right].$$

Since $S_n = \sum_{i=1}^n X_i$ and the Bernoulli variables are independent, then

$$\begin{aligned} M_{S_n}(t) &= \mathbb{E}\left[e^{tX_1}\right] \cdot \mathbb{E}\left[e^{tX_2}\right] \cdots \mathbb{E}\left[e^{tX_n}\right] \\ M_{S_n}(t) &= \prod_{i=1}^n M_{X_i}(t). \end{aligned}$$

From (a), the Bernoulli moment-generating function is $M_{X_i}(t) = 1 - p_i + p_i e^t$. Hence,

$$M_{S_n}(t) = \prod_{i=1}^n \left(1 - p_i + p_i e^t\right).$$

(c) Give a necessary and sufficient condition on $p_1, \ldots, p_n$ under which S_n has a Binomial(n, p) distribution for some p. Justify your answer by comparing MGFs.

Answer:
From (b), the moment-generating function of the sum S_n is

$$M_{S_n}(t) = \prod_{i=1}^n \left(1 - p_i + p_i e^t\right).$$

From the definition of moment-generating function of a Binomial(n, p) distribution,

$$M_{\text{Binomial}}(t) = \left(1 - p + pe^t\right)^n.$$

By the uniqueness theorem of moment-generating function, the sum S_n will follow a Binomial(n, p) distribution if and only if its moment-generating function exactly matches the Binomial moment-generating function for all t in an open interval containing zero. Hence,

$$\prod_{i=1}^{n} \left(1 - p_i + p_i e^t\right) = \left(1 - p + pe^t\right)^n.$$

Let $y = e^t$:

$$\prod_{i=1}^{n} \left[(1 - p_i) + p_i y\right] = \left[(1 - p) + py\right]^n.$$

Then,

$$\left[(1 - p_1) + p_1 y\right] \cdot \left[(1 - p_2) + p_2 y\right] \cdots \left[(1 - p_n) + p_n y\right] = \left[(1 - p) + py\right]^n.$$

By the unique factorization of polynomials, the linear factors must be identical. That is $p_i = p$ for all $i = 1, \ldots, n$. Conversely, if $p_1 = \cdots = p_n = p$, the MGFs match identically. Therefore, $S_n \sim$ Binomial(n, p) if and only if $p_1 = p_2 = \cdots = p_n = p$.

(d) Without assuming that the p_i are equal, derive $\mathbb{E}(S_n)$ and $\text{Var}(S_n)$.

Answer:

For any individual Bernoulli random variable X_i, the expectation is $\mathbb{E}(X_i) = p_i$ and the variance is $\text{Var}(X_i) = p_i(1 - p_i)$.

For the expected value:

Note that by the linearity of expectation, the expected value of a sum equals the sum of the individual expectations. Then,

$$\begin{aligned}
\mathbb{E}(S_n) &= \mathbb{E}\left(\sum_{i=1}^{n} X_i\right) \\
&= \sum_{i=1}^{n} \mathbb{E}(X_i) \\
\mathbb{E}(S_n) &= \sum_{i=1}^{n} p_i.
\end{aligned}$$

For the variance:

Since $X_1, X_2, \ldots, X_n$ are independent, the variance of the sum equals the sum of

the individual variances. Thus,

$$\begin{aligned}\mathrm{Var}(S_n) &= \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \\ &= \sum_{i=1}^{n} \mathrm{Var}(X_i) \\ \mathrm{Var}(S_n) &= \sum_{i=1}^{n} p_i(1-p_i).\end{aligned}$$

(e) Construct two different sets of four probabilities (p_1, p_2, p_3, p_4) that yield the same value of $\mathbb{E}(S_4)$ but different values of $\mathrm{Var}(S_4)$. Verify both claims algebraically.

Answer:
Let the two sets of probabilities be

$$\mathrm{Set}_A = (0.9, 0.5, 0.3, 0.3) \quad \text{and} \quad \mathrm{Set}_B = (0, 1, 0, 1).$$

For Set A:

- The Expected value:

$$\mathbb{E}(S_4) = \sum_{i=1}^{4} p_i = 0.9 + 0.5 + 0.3 + 0.3 = 2.$$

- The Variance:

$$\begin{aligned}\mathrm{Var}(S_4) &= \sum_{i=1}^{4} p_i(1-p_i) \\ &= 0.9(1-0.9) + 0.5(1-0.5) + 0.3(1-0.3) + 0.3(1-0.3) \\ &= 0.09 + 0.25 + 0.21 + 0.21 \\ \mathrm{Var}(S_4) &= 0.76.\end{aligned}$$

For Set B:

- The Expected value:

$$\mathbb{E}(S_4) = \sum_{i=1}^{4} p_i = 0 + 1 + 0 + 1 = 2.$$

- The Variance:

$$\begin{aligned}\mathrm{Var}(S_4) &= \sum_{i=1}^{4} p_i(1-p_i) \\ &= 0(1) + 1(0) + 0(1) + 1(0) \\ \mathrm{Var}(S_4) &= 0.\end{aligned}$$

SInce both sets yield the same expected value $\mathbb{E}(S_4) = 2$, however it have a distinct variances $\mathrm{Var}(S_4)_A = 0.76 \neq 0 = \mathrm{Var}(S_4)_B$.

Problem 3. Joint Distribution on a Nonrectangular Support [16 points]

Suppose the joint probability density function of X and Y is

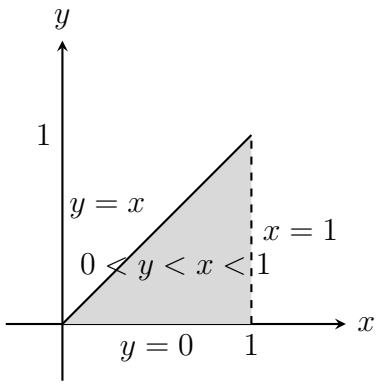
$$f_{X,Y}(x,y) = c(x+y), \quad 0 < y < x < 1,$$

and $f_{X,Y}(x,y) = 0$ otherwise.

(a) Sketch the support of (X,Y) in the xy-plane and label all boundary lines. Your sketch must be consistent with every integral used later in the problem.

Answer:

The support is the region where the joint density is positive $\mathcal{S}_{X,Y} = \{(x,y) : 0 < y < x < 1\}$. This describes a triangular region bounded by $y = 0$, $x = 1$, and $y = x$.



(b) Determine the normalizing constant c.

Answer:

By edefinition, a valid joint probability random function must satisfy $\iint_{\mathcal{S}} f_{X,Y}(x,y)\,dA = 1$. Thus, integrating over $0 < y < x < 1$,

$$\begin{aligned}
1 &= \int_0^1 \int_0^x c(x+y)\,dy\,dx \\
&= c\int_0^1 \left[xy + \frac{y^2}{2}\right]_{y=0}^{y=x} dx \\
&= c\int_0^1 \left(x^2 + \frac{x^2}{2}\right) dx \\
&= c\int_0^1 \frac{3}{2}x^2\,dx \\
&= c\left[\frac{1}{2}x^3\right]_0^1 = c\left(\frac{1}{2}\right).
\end{aligned}$$

Setting $\frac{c}{2} = 1$. Therefore,

$$c = 2$$

(c) Derive the marginal densities $f_X(x)$ and $f_Y(y)$, including their supports.

Answer:
Note that $f_{X,Y}(x, y) = 2(x + y)$ on the nonrectangular support $0 < y < x < 1$ (where $c = 2$ from part b).

By definition, the marginal probability random function of X is

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$$

For $x \in (0, 1)$, the inequality $0 < y < x < 1$. Then,

$$\begin{aligned} f_X(x) &= \int_0^x 2(x + y)\, dy \\ &= 2\left[xy + \frac{y^2}{2}\right]_{y=0}^{y=x} \\ &= 2\left(x(x) + \frac{x^2}{2} - 0\right) \\ &= 2\left(\frac{3}{2}x^2\right) = 3x^2. \end{aligned}$$

Therefore,

$$f_X(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Similarly, the marginal probability random function of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx$$

For $y \in (0, 1)$, the inequality is $0 < y < x < 1$. Then,

$$\begin{aligned} f_Y(y) &= \int_y^1 2(x + y)\, dx \\ &= 2\left[\frac{x^2}{2} + xy\right]_{x=y}^{x=1} \\ &= 2\left[\left(\frac{1^2}{2} + (1)y\right) - \left(\frac{y^2}{2} + y(y)\right)\right] \\ &= 2\left[\frac{1}{2} + y - \frac{3}{2}y^2\right] = 1 + 2y - 3y^2. \end{aligned}$$

Therefore,

$$f_Y(y) = \begin{cases} 1 + 2y - 3y^2, & 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

(d) Derive the conditional density $f_{Y|X}(y \mid x)$ and state its conditional support.

Answer:
From the answer in (c), the conditional density function $f_{Y|X}(y \mid x)$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Then,

$$\begin{aligned} f_{Y|X}(y \mid x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\ &= \frac{2(x+y)}{3x^2} \\ &= \frac{2}{3x} + \frac{2y}{3x^2}, \end{aligned}$$

Therefore,

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{2(x+y)}{3x^2}, & 0 < y < x \\ 0, & \text{otherwise} \end{cases}$$

(e) Derive $\mathbb{E}(Y \mid X = x)$.

Answer:
From the conditional probability density function derived in (d), $f_{Y|X}(y \mid x) = \dfrac{2(x+y)}{3x^2}$ for $0 < y < x$, $\mathbb{E}(Y \mid X = x)$ is

$$\begin{aligned} \mathbb{E}(Y \mid X = x) &= \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x)\, dy \\ &= \int_0^x y \left(\frac{2(x+y)}{3x^2} \right) dy \\ &= \frac{2}{3x^2} \int_0^x (xy + y^2)\, dy \\ &= \frac{2}{3x^2} \left[\frac{xy^2}{2} + \frac{y^3}{3} \right]_{y=0}^{y=x} \\ &= \frac{2}{3x^2} \left(\frac{x(x^2)}{2} + \frac{x^3}{3} - 0 \right) \\ &= \frac{2}{3x^2} \left(\frac{x^3}{2} + \frac{x^3}{3} \right) \\ &= \frac{2}{3x^2} \left(\frac{5x^3}{6} \right) \\ &= \frac{10x^3}{18x^2} = \frac{5}{9}x \end{aligned}$$

Therefore,

$$\mathbb{E}(Y \mid X = x) = \frac{5}{9}x \quad \text{for } 0 < x < 1.$$

(f) Compute $\mathbb{E}(X)$, $\mathbb{E}(Y)$, and $\mathbb{E}(XY)$, and hence determine $\mathrm{Cov}(X, Y)$.

Answer:
For $\mathbb{E}(X)$:
Using the marginal density $f_X(x) = 3x^2$ for $0 < x < 1$, then

$$\begin{aligned}
\mathbb{E}(X) &= \int_0^1 x f_X(x)\, dx \\
&= \int_0^1 x(3x^2)\, dx \\
&= \int_0^1 3x^3\, dx \\
&= \left[\frac{3}{4}x^4\right]_0^1 \\
\mathbb{E}(X) &= \frac{3}{4}
\end{aligned}$$

For $\mathbb{E}(Y)$:
Using the marginal density $f_Y(y) = 1 + 2y - 3y^2$ for $0 < y < 1$, then

$$\begin{aligned}
\mathbb{E}(Y) &= \int_0^1 y f_Y(y)\, dy \\
&= \int_0^1 y(1 + 2y - 3y^2)\, dy \\
&= \int_0^1 (y + 2y^2 - 3y^3)\, dy \\
&= \left[\frac{y^2}{2} + \frac{2y^3}{3} - \frac{3y^4}{4}\right]_0^1 \\
&= \frac{1}{2} + \frac{2}{3} - \frac{3}{4} \\
&= \frac{6 + 8 - 9}{12} \\
\mathbb{E}(Y) &= \frac{5}{12}
\end{aligned}$$

For $\mathbb{E}(XY)$:
Using the joint density $f_{X,Y}(x, y) = 2(x + y)$ over the support $0 < y < x < 1$, then

$$\begin{aligned}
\mathbb{E}(XY) &= \int_0^1 \int_0^x xy \cdot 2(x + y)\, dy\, dx \\
&= 2\int_0^1 \int_0^x (x^2 y + xy^2)\, dy\, dx.
\end{aligned}$$

Evaluating the inner integral first,

$$\begin{aligned}\int_0^x (x^2y + xy^2)\,dy &= \left[\frac{x^2y^2}{2} + \frac{xy^3}{3}\right]_{y=0}^{y=x} \\ &= \frac{x^4}{2} + \frac{x^4}{3} \\ &= \frac{5x^4}{6}\end{aligned}$$

Evaluating the outer integral,

$$\begin{aligned}\mathbb{E}(XY) &= 2\int_0^1 \frac{5x^4}{6}\,dx \\ &= \frac{5}{3}\int_0^1 x^4\,dx \\ &= \frac{5}{3}\left[\frac{x^5}{5}\right]_0^1 \\ \mathbb{E}(XY) &= \frac{1}{3}\end{aligned}$$

For Cov(X, Y):
By definition of covariance,

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Hence,

$$\begin{aligned}\mathrm{Cov}(X, Y) &= \frac{1}{3} - \left(\frac{3}{4}\right)\left(\frac{5}{12}\right) \\ &= \frac{1}{3} - \frac{15}{48} \\ &= \frac{16}{48} - \frac{15}{48} \\ &= \frac{1}{48}\end{aligned}$$

Therefore,

$$\mathrm{Cov}(X, Y) = \frac{1}{48}.$$

(g) Determine whether X and Y are independent. Give two separate arguments: one based on the support and one based on the factorization criterion.

Answer:
For X and Y to be independent, $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$ must hold for all (x, y). Note that from (c),

$$f_X(x) = 3x^2 \quad \text{and} \quad f_Y(y) = 1 + 2y - 3y^2 \quad \text{on } (0, 1).$$

Hence, the product $f_X(x)f_Y(y)$ is

$$f_X(x)f_Y(y) = 3x^2(1 + 2y - 3y^2)$$
$$f_X(x)f_Y(y) = 3x^2 + 6x^2y - 9x^2y^2.$$

Note that,

$$f_{X,Y}(x, y) = 2(x + y) = 2x + 2y.$$

Hence, $3x^2 + 6x^2y - 9x^2y^2 \neq 2x + 2y$.
Moreover, the support $0 < y < x < 1$ is triangular (nonrectangular). Thus, the bounds of y depend on x.
Therefore, X and Y are dependent.

Problem 4. Zero Correlation and Dependence [10 points]

Let

$$X \sim \text{Uniform}(-1, 1),$$

and define

$$Y = X^2.$$

(a) Derive $\mathbb{E}(X)$, $\mathbb{E}(Y)$, and $\mathbb{E}(XY)$ from the distribution of X.

(b) Show that $\text{Cov}(X, Y) = 0$.

(c) Prove that X and Y are not independent. Your proof may not rely solely on the statement "Y is a function of X"; use a probability or conditional-distribution argument.

(d) Explain, in no more than four mathematically precise sentences, why zero correlation fails to imply independence in this example.

Problem 5. A Non-One-to-One Transformation [14 points]

Let

$$X \sim \text{Uniform}(0, 1)$$

and define

$$Y = X(1 - X).$$

(a) Determine the support of Y.

Answer:

Let $X \sim U(0,1)$, then the probability density function of X is

$$f_X(x) = \begin{cases} 1, & \text{if } 0 < x < 1, \\ 0, & \text{if otherwise.} \end{cases}$$

Define $Y = X(1-X)$. Since $0 < x < 1$, then

$$0 < x(1-x) < \frac{1}{4}$$

Therefore,

$$0 < y < \frac{1}{4}$$

(b) Show that the transformation $y = x(1-x)$ is not one-to-one on $(0,1)$, and obtain all inverse branches $x = x(y)$.

Answer:

From

$$\begin{aligned} y &= x(1-x) \\ y &= x - x^2 \end{aligned}$$

Solving for x in terms of y, then

$$\begin{aligned} y &= x - x^2 \\ x^2 - x + y &= 0 \end{aligned}$$

Thus,

$$x = \frac{1 \pm \sqrt{1-4y}}{2}.$$

Therefore, the two inverse branches are

$$x_1(y) = \frac{1+\sqrt{1-4y}}{2} \quad \text{and} \quad x_2(y) = \frac{1-\sqrt{1-4y}}{2}$$

(c) Using the CDF method, derive $F_Y(y)$ for all real y. Your derivation must explicitly account for both inverse branches.

Answer:

From the branches,

$$x < x_1(y) \quad \text{or} \quad x > x_2(y)$$

Then,

$$F_Y(y) = P(x < x_1(y)) + P(x > x_2(y))$$

Since $f_X(x) = 1$ for $0 < x < 1$, then

$$F_Y(y) = x_1(y) + [1 - x_2(y)]$$

Thus substituting the branches,

$$\begin{aligned} F_Y(y) &= \frac{1 + \sqrt{1-4y}}{2} + \left[1 - \frac{1 - \sqrt{1-4y}}{2}\right] \\ &= 1 - \sqrt{1-4y} \end{aligned}$$

Therefore,

$$F_Y(y) = \begin{cases} 0, & y \leq 0 \\ 1 - \sqrt{1-4y}, & 0 < y < \dfrac{1}{4} \\ 1, & y \geq \dfrac{1}{4} \end{cases}$$

(d) Differentiate the CDF to obtain $f_Y(y)$, including its support.

Answer:
Differentiate $F_Y(y)$ with respect to y:

$$\begin{aligned} f_Y(y) &= \frac{d}{dy}\left[1 - \sqrt{1-4y}\right] \\ &= -\frac{1}{2}(1-4y)^{-\frac{1}{2}}(-4) \\ f_Y(y) &= \frac{2}{\sqrt{1-4y}} \end{aligned}$$

Therefore,

$$f_Y(y) = \begin{cases} \dfrac{2}{\sqrt{1-4y}}, & 0 < y < \dfrac{1}{4} \\ 0, & \text{otherwise} \end{cases}$$

(e) Verify independently that your derived PDF integrates to 1.

Answer:
By integrating the probability density function $f_Y(y)$ over its support $\left(0, \frac{1}{4}\right)$:

$$\begin{aligned} \int_0^{\frac{1}{4}} \frac{2}{\sqrt{1-4y}}\, dy &= \left[-\sqrt{1-4y}\right]_0^{\frac{1}{4}} \\ &= -\sqrt{1 - 4\left(\frac{1}{4}\right)} - \left[-\sqrt{1-4(0)}\right] \\ &= -\sqrt{0} - (-\sqrt{1}) \\ \int_0^{\frac{1}{4}} \frac{2}{\sqrt{1-4y}}\, dy &= 1 \end{aligned}$$

Therefore,

$$\int_0^{\frac{1}{4}} \frac{2}{\sqrt{1-4y}}\, dy = 1$$

(f) Compute $\mathbb{E}(Y)$ directly from $\mathbb{E}[g(X)]$. Then compute $\mathbb{E}(Y)$ again using your derived density f_Y and verify that the two answers agree.

Answer:

Let X have probability density function

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Let $Y = g(X) = X(1-X) = X - X^2$.

By the Law of the Unconscious Statistician , the expectation of $Y = g(X) = X(1-X)$ is given by

$$\mathbb{E}(Y) = \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

Then, Evaluating $E(X)$:

$$\begin{aligned} E(X) &= \int_0^1 x(1)\, dx \\ &= \left[\frac{x^2}{2}\right]_0^1 \\ &= \frac{1^2}{2} - \frac{0^2}{2} \\ &= \frac{1}{2} \end{aligned}$$

Evaluating $E(X^2)$:

$$\begin{aligned} E(X^2) &= \int_0^1 x^2(1)\, dx \\ &= \left[\frac{x^3}{3}\right]_0^1 \\ &= \frac{1^3}{3} - \frac{0^3}{3} \\ &= \frac{1}{3} \end{aligned}$$

From the Linearity of Expectation,

$$E(X - X^2) = E(X) - E(X^2).$$

Then,

$$E(X - X^2) = \frac{1}{2} - \frac{1}{3}$$
$$E(X - X^2) = \frac{1}{6}$$

Let Y have the derived probability density function

$$f_Y(y) = \begin{cases} \dfrac{2}{\sqrt{1-4y}}, & 0 < y < \dfrac{1}{4}, \\ 0, & \text{elsewhere.} \end{cases}$$

From the definition of Expected value of a continuous random variable,

$$\mu = E(Y) = \int_{-\infty}^{\infty} y f_Y(y)\, dy$$

Then,

$$E(Y) = \int_0^{\frac{1}{4}} y \left(\frac{2}{\sqrt{1-4y}} \right) dy$$

Let $u = \sqrt{1-4y}$ (where $u^2 = 1 - 4y \implies y = \frac{1-u^2}{4}$ and $du = -\frac{2}{\sqrt{1-4y}}\, dy \implies \left(\frac{2}{\sqrt{1-4y}}\right) dy = -du$), with bounds $y = 0 \to u = 1$ and $y = \frac{1}{4} \to u = 0$. Then,

$$E(Y) = \int_1^0 \left(\frac{1-u^2}{4} \right) (-du)$$

Thus, by reversing the bounds of integration,

$$\begin{aligned} E(Y) &= \int_0^1 \frac{1-u^2}{4}\, du \\ &= \frac{1}{4} \left[u - \frac{u^3}{3} \right]_0^1 \\ &= \frac{1}{4} \left(1 - \frac{1}{3} \right) - \frac{1}{4} \left(0 - \frac{0}{3} \right) \\ &= \frac{1}{4} \left(\frac{2}{3} \right) \\ &= \frac{1}{6} \end{aligned}$$

Since $E(Y) = \dfrac{1}{6}$. Therefore, the two methods of computing $\mathbb{E}(Y)$ agree.

Problem 6. Multivariate Transformation and Jacobian [16 points]

Let X and Y be independent random variables with densities

$$f_X(x) = \frac{x^{\alpha-1}e^{-x}}{\Gamma(\alpha)}, \quad x > 0,$$

$$f_Y(y) = \frac{y^{\beta-1}e^{-y}}{\Gamma(\beta)}, \quad y > 0,$$

where $\alpha > 0$ and $\beta > 0$. Define the transformation

$$U = X + Y, \quad V = \frac{X}{X+Y}.$$

(a) Solve for x and y as functions of u and v.

Answer:
Define the transformation

$$U = X + Y, \quad V = \frac{X}{X+Y}.$$

Then,

$$V = \frac{X}{X+Y}$$

Since $U = X + Y$, then

$$v = \frac{x}{u}$$
$$x = uv$$

From $U = X + Y$, substituting $x = uv$ then

$$u = uv + y$$

Hence,

$$y = u - uv$$
$$y = u(1 - v)$$

Therefore,

$$x(u, v) = uv \quad \text{and} \quad y(u, v) = u(1 - v).$$

(b) Determine the transformed support of (U, V) directly from $x > 0$ and $y > 0$.

Answer:
From (a), the inverse transformations are $x = uv$ and $y = u(1 - v)$.
Since $U = X + Y$ and $X > 0$ and $Y > 0$, then

$$u = x + y > 0$$

From $x > 0$, substitute $x = uv$. Thus,

$$\begin{aligned} x > 0 &\implies uv > 0 \\ &\implies \frac{1}{u}(uv > 0) \\ &\implies v > 0 \end{aligned}$$

From $y > 0$, substitute $y = u(1 - v)$. Then,

$$\begin{aligned} y > 0 &\implies u(1 - v) > 0 \\ &\implies \frac{1}{u}(u(1 - v) > 0) \\ &\implies 1 - v > 0 \\ &\implies v < 1 \end{aligned}$$

Therefore,

$$u > 0 \quad \text{and} \quad 0 < v < 1.$$

(c) Compute the Jacobian determinant

$$\left|\frac{\partial(x, y)}{\partial(u, v)}\right|$$

from the derivative matrix.

Answer:
Note that from (a), $x(u, v) = uv$ and $y(u, v) = u(1 - v)$. Then,

$$\begin{pmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{pmatrix} = \begin{pmatrix} v & u \\ 1 - v & -u \end{pmatrix}$$

From the Jacobian determinant $J = \dfrac{\partial(x, y)}{\partial(u, v)}$, then

$$\begin{aligned} J &= \det \begin{pmatrix} v & u \\ 1 - v & -u \end{pmatrix} \\ &= (v)(-u) - (u)(1 - v) \\ &= -uv - u + uv \\ J &= -u \end{aligned}$$

Therefore,

$$\begin{aligned} |J| &= |-u| \\ |J| &= u \end{aligned}$$

(d) Derive the joint density $f_{U,V}(u,v)$, including the support.

Answer:
Since X and Y are independent random variables, the joint probability density function is

$$f_{X,Y}(x,y) = f_X(x)\cdot f_Y(y) = \frac{x^{\alpha-1}e^{-x}}{\Gamma(\alpha)}\cdot\frac{y^{\beta-1}e^{-y}}{\Gamma(\beta)} = \frac{x^{\alpha-1}y^{\beta-1}e^{-(x+y)}}{\Gamma(\alpha)\Gamma(\beta)} \quad \text{for } x>0, y>0$$

Hence, the joint probability density function of (U,V) is

$$f_{U,V}(u,v) = f_{X,Y}\big(x(u,v), y(u,v)\big)\cdot|J|$$

Note that from (a), $x = uv$ and $y = u(1-v)$. Then,

$$\begin{aligned} f_{X,Y}(uv, u(1-v)) &= \frac{x^{\alpha-1}y^{\beta-1}e^{-(x+y)}}{\Gamma(\alpha)\Gamma(\beta)}\cdot|J| \\ &= \frac{(uv)^{\alpha-1}[u(1-v)]^{\beta-1}e^{-[uv+u(1-v)]}}{\Gamma(\alpha)\Gamma(\beta)}\cdot|J| \\ &= \frac{u^{\alpha+\beta-2}v^{\alpha-1}(1-v)^{\beta-1}e^{-u}}{\Gamma(\alpha)\Gamma(\beta)}\cdot|J| \\ f_{X,Y}(uv, u(1-v)) &= \frac{u^{\alpha+\beta-2}v^{\alpha-1}(1-v)^{\beta-1}e^{-u}}{\Gamma(\alpha)\Gamma(\beta)}\cdot|J| \end{aligned}$$

From (c), the $|J| = u$. Then,

$$\begin{aligned} f_{U,V}(u,v) &= \left(\frac{u^{\alpha+\beta-2}v^{\alpha-1}(1-v)^{\beta-1}e^{-u}}{\Gamma(\alpha)\Gamma(\beta)}\right)\cdot u \\ &= \frac{u^{\alpha+\beta-1}e^{-u}}{\Gamma(\alpha)\Gamma(\beta)}\,v^{\alpha-1}(1-v)^{\beta-1} \end{aligned}$$

Therefore,

$$f_{U,V}(u,v) = \begin{cases} \dfrac{u^{\alpha+\beta-1}e^{-u}}{\Gamma(\alpha)\Gamma(\beta)}\,v^{\alpha-1}(1-v)^{\beta-1}, & u>0 \text{ and } 0<v<1 \\ 0, & \text{otherwise} \end{cases}$$

(e) Derive the marginal density of U and identify its distribution.

(f) Derive the marginal density of V and identify its distribution.

(g) Determine whether U and V are independent. Justify your conclusion by factorizing the joint density, not by inspection alone.

(h) Specialize your result to $\alpha = \beta = 1$. What is the distribution of V in this case? Explain why this agrees with your general result.

Problem 7. Order Statistics from First Principles [14 points]

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed continuous random variables with common CDF F and PDF f. Define

$$X_{(1)} = \min(X_1, \ldots, X_n), \quad X_{(n)} = \max(X_1, \ldots, X_n).$$

(a) Starting from the event $\{X_{(n)} \le x\}$, derive the CDF and PDF of $X_{(n)}$. Do not quote a memorized order-statistic formula.

(b) Starting from the event $\{X_{(1)} > x\}$, derive the CDF and PDF of $X_{(1)}$.

(c) For $a < b$, derive

$$P(X_{(1)} > a, X_{(n)} < b)$$

in terms of $F(a)$ and $F(b)$. State any continuity assumptions you use.

(d) Now suppose $X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$, where $\theta > 0$. Derive the PDFs of $X_{(1)}$ and $X_{(n)}$ explicitly.

(e) For the Uniform$(0, \theta)$ case, derive $\mathbb{E}[X_{(n)}]$ from its density.

(f) Find a constant k_n, depending only on n, such that

$$\mathbb{E}[k_n X_{(n)}] = \theta.$$

Derive k_n rather than guessing it.

Problem 8. Proof Critique and Repair [6 points]

Consider the following argument about two random variables X and Y:

> "Since $\text{Cov}(X, Y) = 0$, X and Y are independent. Therefore
>
> $$f_{X,Y}(x, y) = f_X(x) f_Y(y),$$
>
> and consequently
>
> $$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$
>
> "

(a) Identify the first logically invalid implication in the argument and explain precisely why it is invalid.

(b) Using the random variables in Problem 4, provide a counterexample to the invalid implication.

(c) State a correct sufficient condition under which

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

holds.

(d) Rewrite the original argument as a logically valid implication chain under an appropriate assumption.